EXPERIMENT 20:Prolog Program for Planets Database

Aim

To write and implement a Prolog program to create a database of planets and retrieve information about planets using facts and rules.

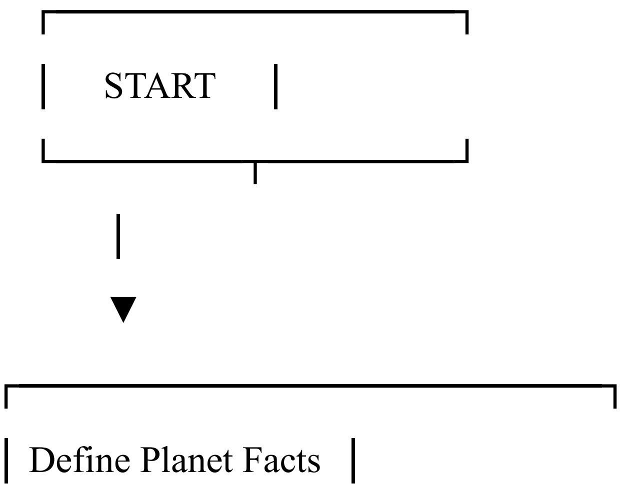
Objective

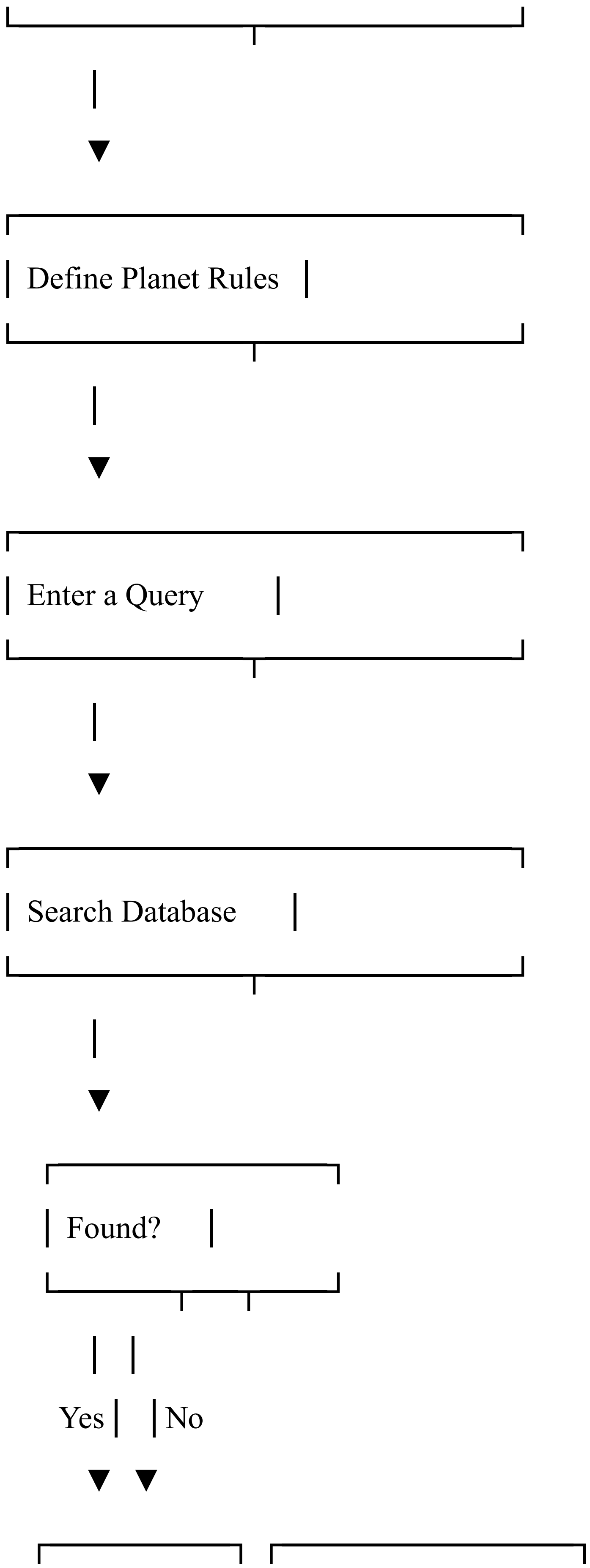
- To understand facts and rules in Prolog.
- To store information about planets.
- To retrieve planet details using queries.
- To determine properties such as planet type and position.

Algorithm

5. Start the program.
6. Define facts for planets in the Solar System.
7. Store information such as planet name, position, and type.
8. Define rules to identify terrestrial and gas-giant planets.
9. Execute queries to retrieve planet information.
10. Display the result.
11. Stop the program.

Flowchart




Define Planet Rules
Enter a Query
Search Database
Found?
Yes
No

```
| Display |  | Display      |
| Result  |  | "Not Found"  |
└────┬─────┘  └──────┬──────────┘
  |         |
  └────────┬────────┘
       ▼
  ┌────────────────┐
  |   END   |
  └────────────────┘
```

Code

```
planets = [
    {
        "name": "Mercury",
        "type": "Terrestrial",
        "distance_from_sun": 57.9,
        "moons": 0
    },
    {
        "name": "Venus",
        "type": "Terrestrial",
        "distance_from_sun": 108.2,
        "moons": 0
    },
    {
        "name": "Earth",
```

"type": "Terrestrial",
"distance_from_sun": 149.6,
"moons": 1
},
{
"name": "Mars",
"type": "Terrestrial",
"distance_from_sun": 227.9,
"moons": 2
},
{
"name": "Jupiter",
"type": "Gas Giant",
"distance_from_sun": 778.5,
"moons": 95
},
{
"name": "Saturn",
"type": "Gas Giant",
"distance_from_sun": 1434.0,
"moons": 146
},
{
"name": "Uranus",
"type": "Ice Giant",

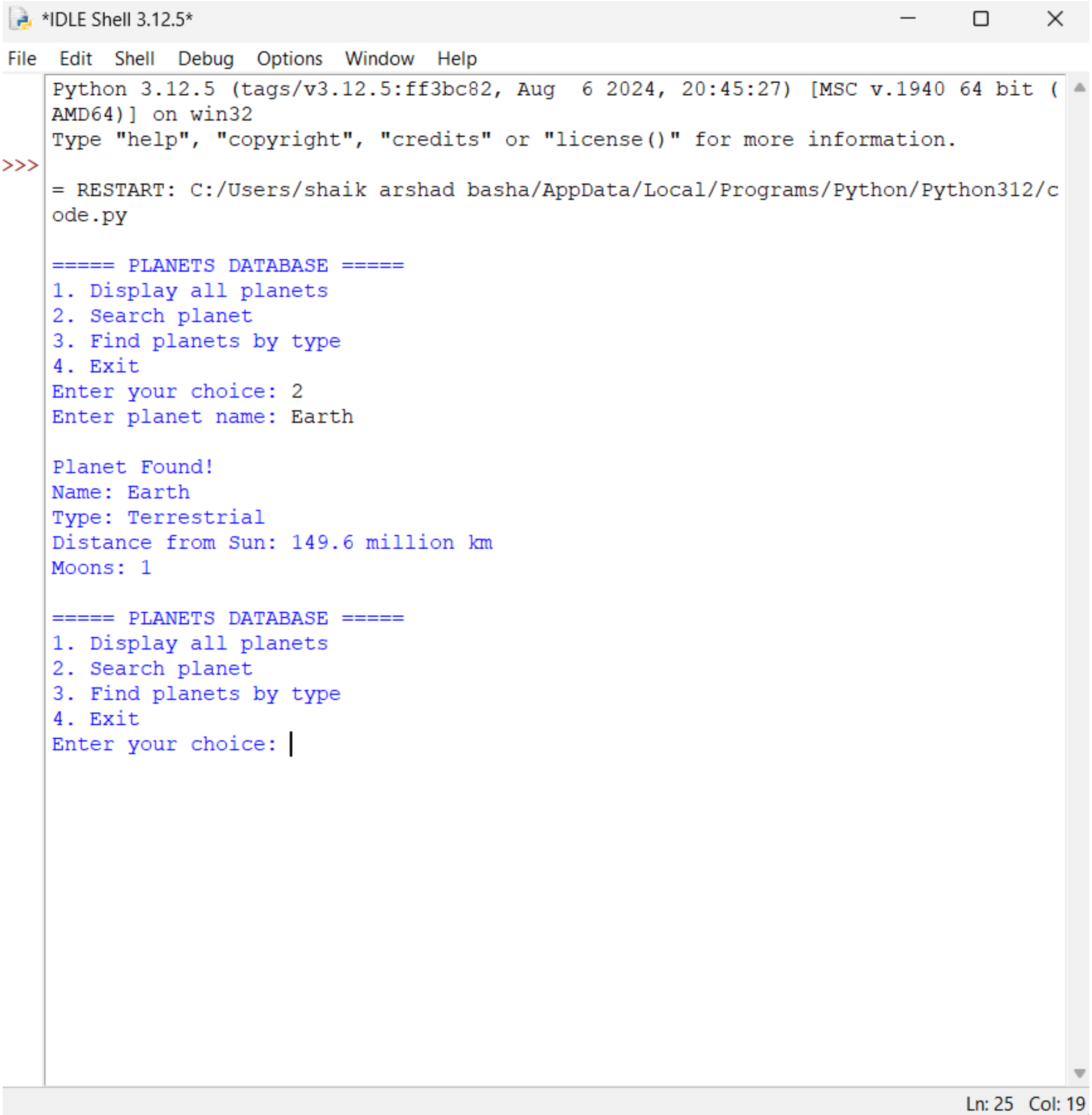
```
        "distance_from_sun": 2871.0,
        "moons": 28
    },
    {
        "name": "Neptune",
        "type": "Ice Giant",
        "distance_from_sun": 4495.0,
        "moons": 16
    }
]
def display_planets():
    print("\n--- PLANETS DATABASE ---")
    for planet in planets:
        print("Name:", planet["name"])
        print("Type:", planet["type"])
        print("Distance from Sun:", planet["distance_from_sun"], "million km")
        print("Moons:", planet["moons"])
        print("------------------------")
def search_planet(name):
    for planet in planets:
        if planet["name"].lower() == name.lower():
            print("\nPlanet Found!")
            print("Name:", planet["name"])
            print("Type:", planet["type"])
            print("Distance from Sun:",
```

```
            planet["distance_from_sun"], "million km")
        print("Moons:", planet["moons"])
        return
    print("\nPlanet not found.")
def find_by_type(planet_type):
    print("\nPlanets of type:", planet_type)
    found = False
    for planet in planets:
        if planet["type"].lower() == planet_type.lower():
            print("-", planet["name"])
            found = True
    if not found:
        print("No planets found.")
while True:
    print("\n===== PLANETS DATABASE =====")
    print("1. Display all planets")
    print("2. Search planet")
    print("3. Find planets by type")
    print("4. Exit")
    choice = input("Enter your choice: ")
    if choice == "1":
        display_planets()

    elif choice == "2":
        name = input("Enter planet name: ")
```

```
        search_planet(name)
    elif choice == "3":
        planet_type = input(
            "Enter type (Terrestrial / Gas Giant / Ice Giant): "
        )
        find_by_type(planet_type)
    elif choice == "4":
        print("Program terminated.")
        break
    else:
        print("Invalid choice.")
```

Output



Result

The Prolog program was successfully implemented to create a Planets Database and retrieve information about planets using facts and rules.

Conclusion

The program demonstrates how Prolog facts, rules, variables, and queries can be used to create and search a simple knowledge database containing information about planets.